Demand-Responsive Electric Vehicle Routing Under Uncertainty

Independent Research Project

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Abstract

This project investigates routing optimization for demand-responsive electric mobility systems under uncertain demand.

A baseline vehicle routing model was implemented using OR-Tools and evaluated under battery constraints. The results show that routes optimized without considering energy limitations are often infeasible in electric vehicle settings.

To address this, a charging-aware route repair heuristic was developed. The approach dynamically inserts charging stops based on energy requirements and safety constraints.

The model was evaluated across multiple demand scenarios, demonstrating that the proposed method significantly improves feasibility, achieving feasible routes in all tested scenarios. The findings highlight the importance of integrating charging decisions into routing optimization under uncertainty.

Overall, these results highlight the importance of incorporating energy constraints directly into routing optimization models.

1. Introduction

The transition to electric mobility introduces new operational challenges for transportation systems. In particular, electric vehicles are subject to battery limitations, which significantly affect routing decisions.

Demand-responsive mobility systems, such as on-demand shuttle services, must operate efficiently under uncertain demand while satisfying operational constraints. Traditional vehicle routing models typically ignore energy limitations, which may result in solutions that are not feasible in real-world electric vehicle settings.

This project aims to explore the impact of battery constraints and demand uncertainty on routing decisions. In particular, we investigate whether classical routing solutions remain feasible and propose a simple method to improve feasibility by incorporating charging decisions.

The main contribution of this project is to demonstrate how simple heuristic adjustments can restore feasibility in electric vehicle routing under uncertainty, highlighting the need for integrated optimization approaches.

2. Problem Description

We consider a synthetic demand-responsive mobility network consisting of a depot, a charging station, and a set of customer locations distributed in a two-dimensional space.

The objective is to construct a route that starts and ends at the depot while visiting all active customers. The routing problem is solved using a classical vehicle routing approach.

In addition to routing, we introduce energy constraints corresponding to an electric vehicle with limited battery capacity. Each movement between locations consumes energy proportional to the travel distance.

Furthermore, we consider uncertainty in customer demand. In each scenario, only a subset of customers is active, representing real-world variability in demand.

Figure 1 illustrates an example of the generated mobility network and routing structure.

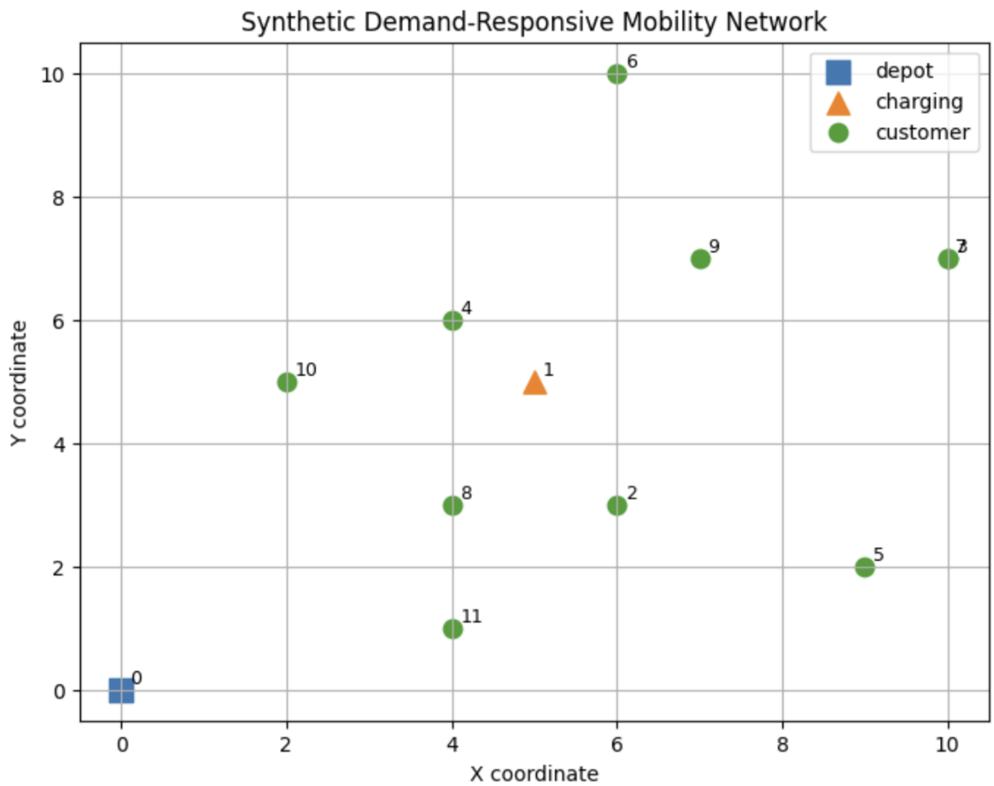

Figure 1. Synthetic mobility network with depot, charging station, and customer locations

3. Methodology

3.1 Baseline Routing Model

A baseline routing model was implemented using Google OR-Tools. The model computes a route that minimizes total travel distance while visiting all required locations.

At this stage, battery constraints are not included in the optimization model. The resulting route is therefore optimal in terms of distance but may not be operationally feasible for electric vehicles.

3.2 Battery Feasibility Evaluation

To evaluate the feasibility of the computed routes, a post-processing step was introduced. The remaining battery level is tracked along the route, assuming a fixed energy consumption rate per unit distance.

A route is considered infeasible if the battery level becomes negative at any point during execution.

3.3 Charging-Aware Route Repair

To improve feasibility, a charging-aware route repair heuristic was developed.

The method iteratively evaluates whether the vehicle can safely reach the next customer and subsequently a safe location (either the depot or a charging station). If not, a charging stop is inserted before proceeding.

After visiting a charging station, the battery is reset to full capacity. This process ensures that the vehicle does not become stranded due to insufficient energy.

Although this approach does not guarantee global optimality, it provides a practical and computationally efficient method for incorporating charging considerations.

3.4 Demand Uncertainty Scenarios

To model demand uncertainty, multiple scenarios were generated in which only a subset of customers is active. Each customer is included in a scenario with a fixed probability.

For each scenario, both the baseline route and the charging-aware repaired route were evaluated in terms of feasibility and route length.

4. Results

This section presents the results of the baseline routing model and the proposed charging-aware repair heuristic.

4.1 Baseline Routing Performance

The baseline routing approach consistently produces infeasible solutions when battery constraints are taken into account.
In all tested scenarios, the vehicle runs out of energy before completing the route.

Figure 2 illustrates the baseline route obtained without considering battery constraints. Although the route is distance-efficient, it does not account for energy limitations and is therefore not operationally feasible.

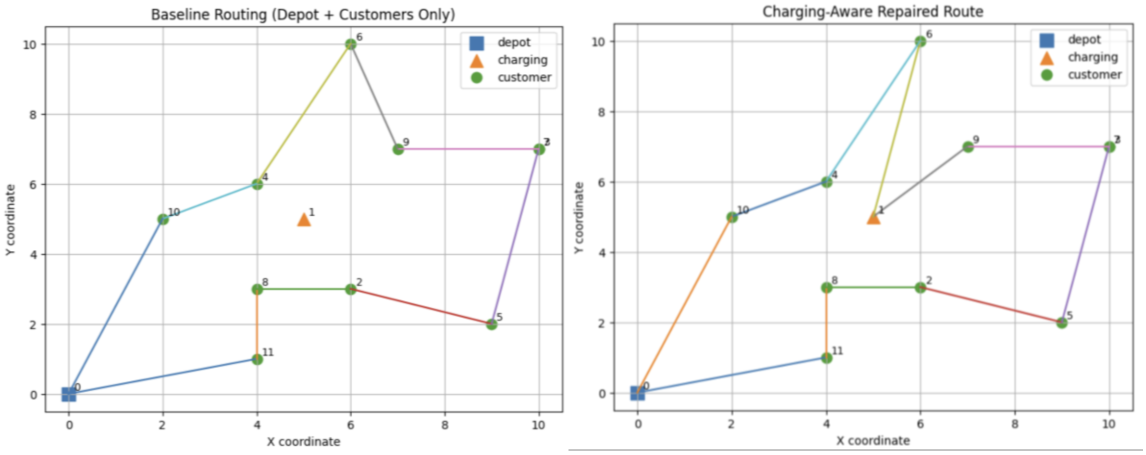

Figure 2. Comparison of baseline routing and charging-aware repaired route

4.2 Charging-Aware Routing

To address this issue, a charging-aware repair heuristic was applied.

Figure 2 also shows the modified route after inserting charging stops. The adjusted route includes visits to the charging station, ensuring that the vehicle maintains sufficient energy throughout the trip.

As a result, the charging-aware approach produces feasible routes in all evaluated scenarios.

4.3 Battery Dynamics

The difference between the two approaches is further illustrated by the battery level evolution.

Figure 3 shows the battery level along the baseline route, where the battery becomes negative before the route is completed.

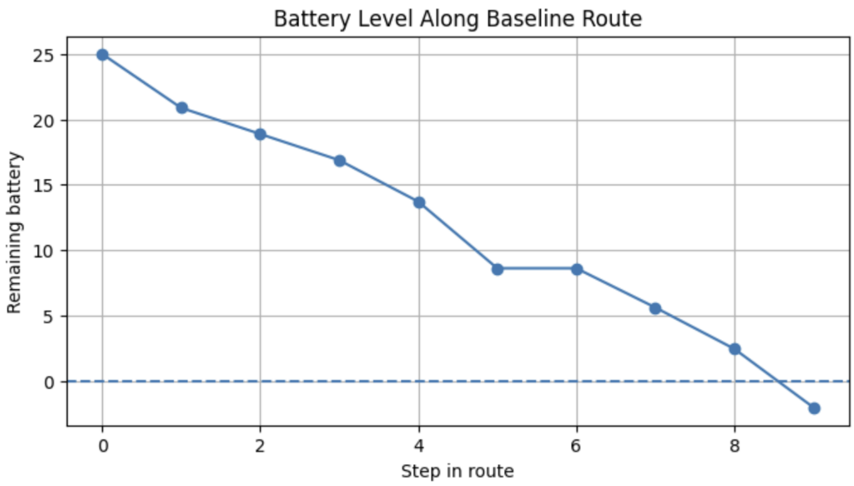

Figure 3. Battery level along baseline route

In contrast, *Figure 4* shows that the charging-aware route maintains a positive battery level throughout the entire trip due to timely recharging.

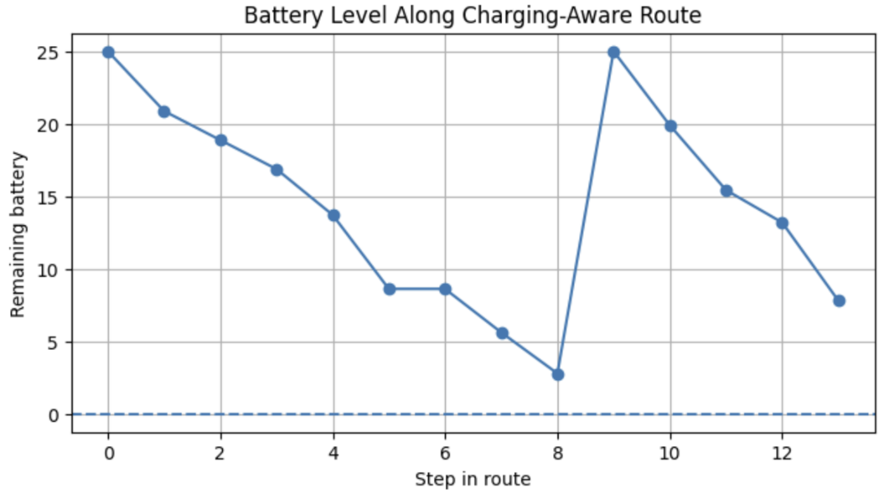

Figure 4. Battery level along charging-aware route

These results clearly demonstrate the importance of incorporating energy constraints into routing decisions.

4.4 Feasibility Comparison

The feasibility of both approaches across all scenarios is summarized in *Table 1*, highlighting the clear advantage of the charging-aware method.

Model	Feasible Scenarios
Baseline	0
Charging-aware	5

Table 1. Feasibility results across scenarios

The charging-aware approach improves feasibility from 0% to 100% across all evaluated scenarios.
This demonstrates that even a simple heuristic adjustment can significantly improve operational feasibility.

4.5 Trade-off Between Feasibility and Efficiency

However, the improved feasibility comes at the cost of longer routes.

Figure 5 shows the number of route steps across different demand scenarios. The charging-aware routes are consistently longer than the baseline routes due to additional detours for charging.

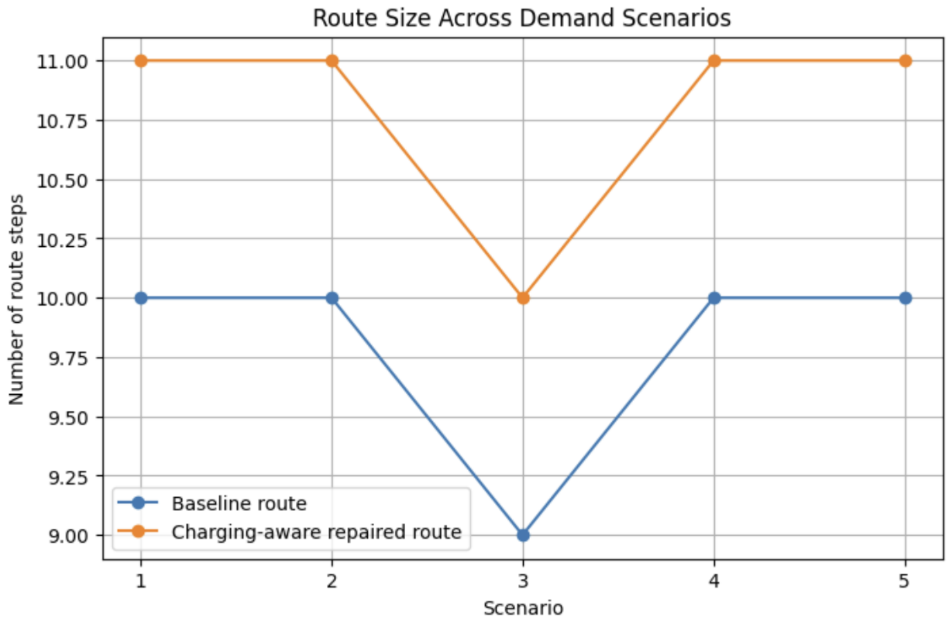

Figure 5. Route size across demand scenarios

These findings highlight a fundamental trade-off between efficiency (shorter routes) and feasibility (energy constraints), which is central to electric vehicle routing problems.

4.6 Impact of Demand Uncertainty

Additionally, the results indicate that uncertainty in customer demand leads to variability in route structure and energy requirements.

Different demand scenarios result in different routing patterns and charging needs, highlighting the importance of adaptive routing strategies in real-world applications.

5. Discussion

The findings suggest that routing decisions in electric mobility systems cannot be separated from energy management. Traditional routing models fail to capture critical operational constraints, leading to infeasible solutions.

The proposed repair heuristic demonstrates that incorporating charging decisions can restore feasibility, but it also highlights the limitations of post-processing approaches.

A more advanced approach would integrate charging decisions directly into the routing optimization model, potentially leading to more efficient and robust solutions.

Furthermore, demand uncertainty introduces additional complexity, as different scenarios may require different routing and charging strategies.

These results suggest that solving electric vehicle routing problems requires integrated optimization approaches rather than sequential or post-processing methods, especially under uncertainty.

6. Conclusion

This project demonstrates that battery constraints and demand uncertainty fundamentally reshape routing decisions in electric mobility systems.

Baseline routing approaches are insufficient in this context, as they often produce infeasible solutions. A simple charging-aware repair heuristic can restore feasibility but introduces additional operational costs.

Future work could focus on integrating energy constraints directly into optimization models and developing adaptive strategies that account for uncertainty in demand and system conditions.

This work provides a simple yet illustrative example of how operational constraints and uncertainty interact in modern mobility systems.